

WorldCrafter: Consistent Video World Model with Implicit 3D-aware Memory

Wangbo Yu^{1*} Kunhao Liu^{1*} Wenbo Hu^{1†} Shenghai Yuan² Haiyang Zhou²
 Yukun Huang¹ Yiran Wang¹ Wang Zhao¹ Yingmin Luo¹ Ying Shan¹
¹ARC Lab, Tencent IEG ²Peking University

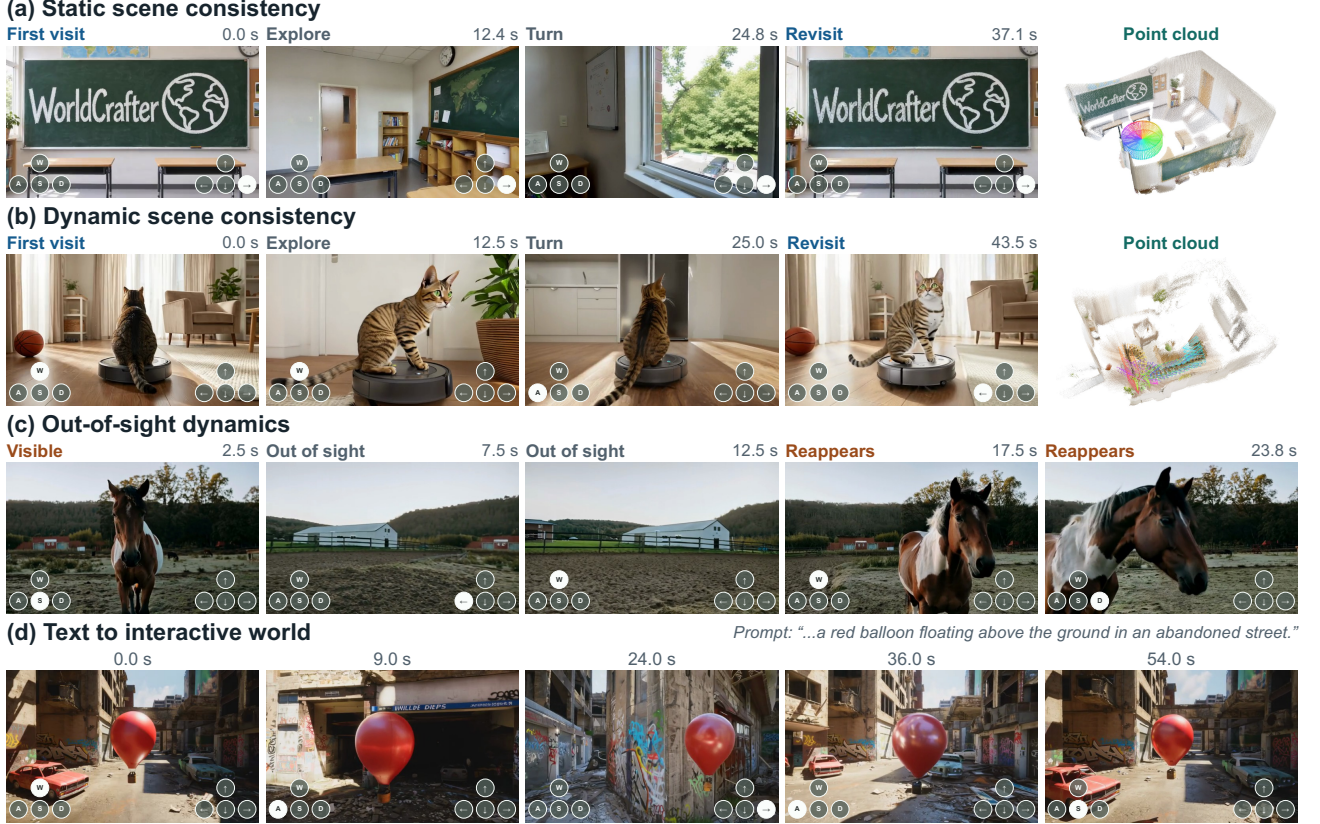


Figure 1. WorldCrafter supports interactive world exploration with camera control. It preserves scene consistency during exploration and revisiting in both static (a) and dynamic (b) scenes, with point clouds reconstructed from the generated videos further illustrating spatial consistency. It also maintains subject continuity across out-of-sight intervals (c) and allows users to explore scenes generated solely from text descriptions (d).

Abstract

Video world models simulate dynamic environment by generating observations in response to user actions, yet preserving previously observed content over long horizons remains challenging. We present WorldCrafter, a video world model that learns a camera-queryable implicit 3D-aware memory for this purpose. The key insight is to let the re-

quested viewpoint shape how multi-view evidence is compressed into the video generator’s limited token budget. Trained jointly with the video generator, a memory encoder and pose-conditioned readout module integrate historical observations into a fixed set of target view-specific tokens before denoising, without explicit depth-based correspondences. By combining this memory with recent temporal context and few-step distillation, WorldCrafter enables streaming scene exploration from a single input image or text prompt. Experiments across static and dynamic scenes

*Equal contribution.

†Corresponding author.

demonstrate substantial improvements in long-horizon consistency and camera-control accuracy while preserving visual quality throughout minute-scale exploration.

1. Introduction

Video world models enable interactive exploration of dynamic environments by generating new observations as users move the camera [5, 11, 14, 29, 39, 42, 51, 59, 83]. Across these developments, memory is essential for grounding successive predictions in a consistent world. It must retain previously observed content beyond the current context and recover it when users revisit explored regions.

A straightforward way to preserve memory for video world models is to involve their previously generated frames in attention computing. Full-history attention incurs substantial computation costs [6, 17, 67], while selective history retrieval trades view coverage for efficiency [6, 8, 31, 48, 49, 59, 61, 63, 67, 70, 72]. Explicit spatial memories provide a shared 3D reference but depend on accurate geometry and struggle with dynamic scenes [62, 69, 75, 82]. Implicit memories compress history into learned representations [44, 47, 72, 76], with recent methods incorporating geometry features for 3D awareness [19, 54, 60]. However, aligning feature spaces learned for geometry estimation with the token space of video generators remains challenging.

Recent advances in 3D representation learning [24–26, 50] have demonstrated remarkable capabilities in learning compact scene representations, making them a natural foundation for the memory space of video world models. Motivated by this, we present **WorldCrafter**, a video world model with implicit 3D-aware memory for consistent interactive generation. At its core, a memory encoder initialized from pretrained 3D representation encoders maps historical latent frames into a compact memory space, inheriting their learned 3D inductive bias. We further optimize this memory space by jointly training the memory encoder, video diffusion transformer (DiT), and a pose-conditioned memory readout module, enabling the memory to co-adapt with the DiT token space and support long-horizon world modeling.

During interaction, we select complementary historical views according to their joint camera coverage and combine the queried memory with recent temporal context. The memory supplies historical scene information, while recent context supports the continuation of visible motion. Each generated chunk is added to the history archive for future queries, while the number of encoder input views and the DiT memory-token budget remain fixed at each step. Together with camera-conditioned autoregressive generation and few-step distillation, our model achieves real-time streaming inference while preserving precise camera control and minute-scale consistency under complex user-

specified camera trajectories.

Our contributions are as follows:

- We introduce an implicit 3D-aware memory mechanism for video world models. It learns to encode history latent frames into a compact memory representation that preserves spatial-temporal context, enabling efficient memory writing and readout within a fixed token budget.
- We integrate this memory mechanism into camera-controllable autoregressive video generation, achieving leading revisit consistency and camera-control accuracy on our benchmark, with controlled ablations supporting our design.
- We build a real-time interactive system through few-step distillation, achieving streaming inference while maintaining visual quality throughout minute-scale exploration.

2. Related Work

2.1. Interactive Video World Models

Video world models generate future observations in response to user actions, turning video generation into interactive simulation [20]. Recent methods increasingly combine streaming generation with camera control to support real-time interaction and long-horizon exploration [1, 5, 10, 12–14, 23, 39, 40, 46, 48, 49, 59, 69, 81, 83].

Streaming generation methods broadly follow two paradigms. The first restructures the denoising process for causal rollout using moving FIFO queues or temporally varying noise levels [7, 27], whereas the second adapts pretrained bidirectional teachers through causal distillation and self-forcing training [9, 21, 38, 65, 68]. Despite their differences, both paradigms autoregressively generate fixed-length chunks and reuse recent outputs as context [16]. Such recent context cannot recover content that has remained outside the context window. Consequently, existing streaming generation methods lack a persistent world state for maintaining long-range consistency.

Camera control is commonly implemented through discrete action inputs [11, 29, 59, 71], continuous camera parameters [3, 15, 32, 57, 78], or point-cloud renders along the target trajectory [45, 73, 74]. Although these signals specify viewpoint changes, they do not provide a persistent state that preserves scene content beyond the context window. Thus, combining streaming generation with camera control alone remains insufficient for consistent long-horizon exploration.

2.2. Memory Mechanisms in Video World Models

Persistent video world models require memory beyond the recent video context. We categorize existing approaches according to their stored representations: context memory, spatial memory, and implicit memory.

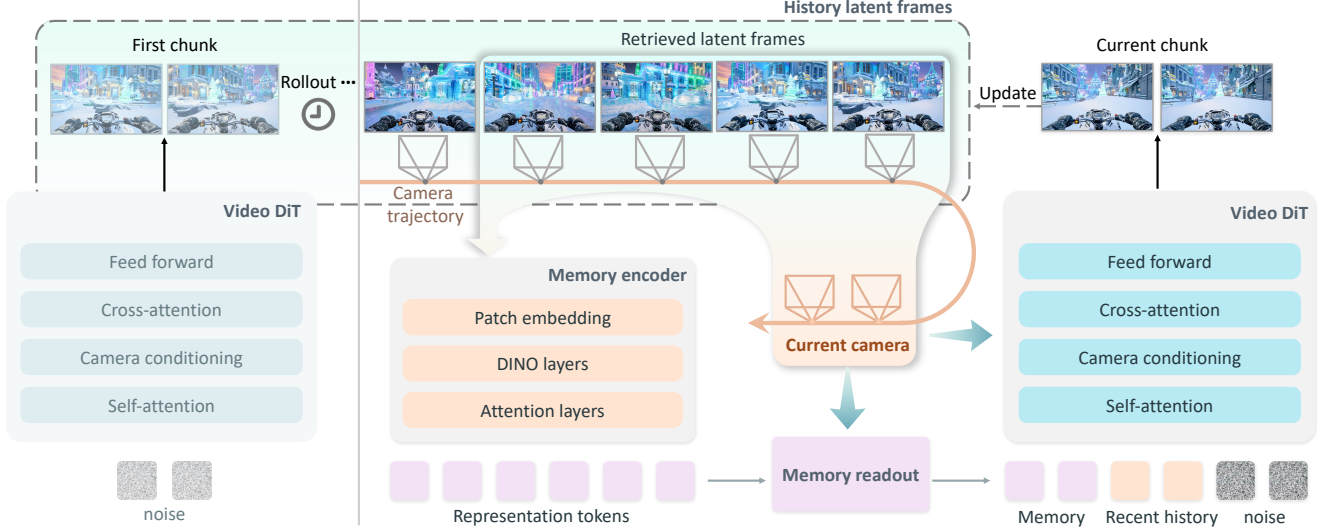


Figure 2. **Overview of the WorldCrafter pipeline.** The video DiT generates the first chunk conditioned on camera poses, before any history is available. At subsequent rollout steps, max-coverage history retrieval selects latent frames based on the current camera poses, and the memory encoder maps them into a compact 3D-aware representation. A pose-conditioned memory readout module extracts fixed-size memory tokens that, together with recent history, guide generation of the current chunk. Generated frames are added to the history for subsequent rollouts.

Context memory retains historical frames or their latent tokens for attention-based reuse [6, 17, 41, 49, 59, 63, 70, 79]. Frame-retrieval methods select a small subset using camera overlap or reconstructed surfaces [18, 31, 70], while learned querying can aggregate information across the available history. MemLearner [72] uses query tokens that attend to both historical context and noisy predictions in shallow DiT layers; deeper layers consume the queries without the original context tokens. CaR [44] compresses historical latents and retrieves them through relative-camera attention inside the denoising network. Both adapt memory access within the generator, whereas WorldCrafter forms its memory conditions through a separate target-pose readout before denoising.

Spatial memory instead transforms history frames into views specified by target camera poses [30, 45, 58, 62, 64, 69, 74, 75, 82]. Representative methods perform this transformation via novel view synthesis, encode the synthesized target-view frames into the VAE latent space, and concatenate the resulting latents channel-wise with the input noise to condition video diffusion models [30, 45, 58, 62, 74]. The resulting spatial correspondence across viewpoints facilitates revisiting previously observed regions. However, strong alignment to the target view can overconstrain scene dynamics, limiting their ability to model dynamic objects and environments.

Implicit memory encodes history into learned representations [8, 33, 47, 61, 76]. Existing methods update memory recurrently alongside local context [47, 76] or compress observations using learned encoders [8, 61]. Geometry-

aware approaches use features from pretrained models such as VGGT [54] as generation conditions or memory supervision [19, 60]. GIM-World [60] jointly trains its memory encoder and generator, using a camera-queryable head to supervise the memory with geometric features; this head is discarded at inference. Its memory tokens summarize history without future-camera queries, while WorldCrafter uses the requested target poses to read out the conditions supplied to the generator during inference. We initialize the multi-view encoder and readout from LagerNVS [50] and adapt them jointly with video generation. Our distinction concerns how target-conditioned memory is formed and used, rather than joint training or implicit memory alone.

3. Method

3.1. Preliminary

Latent video diffusion. Let $\mathbf{x} \in \mathbb{R}^{3 \times F \times H \times W}$ denote a clean video of F frames. A video variational autoencoder (VAE) [28] maps it to a spatiotemporal latent $\mathbf{z} = \mathcal{E}_{\text{VAE}}(\mathbf{x}) \in \mathbb{R}^{c \times f \times h \times w}$. Its decoder reconstructs the video as $\hat{\mathbf{x}} = \mathcal{D}_{\text{VAE}}(\mathbf{z})$. Operating in this latent space substantially reduces the token sequence processed by the Diffusion Transformer (DiT) [43]-based denoiser, which patchifies $\mathbf{z}$ into video tokens and applies 3D self-attention. In a conventional bidirectional video diffusion model such as Wan 2.1 [53], every video token can attend to all other tokens in the clip during each denoising step.

The denoiser is trained with conditional flow matching [37]. For diffusion time $t \sim \mathcal{U}(0, 1)$ and Gaussian noise

$\epsilon \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$, the noisy latent and its target velocity are

$$\mathbf{z}_t = (1 - t)\mathbf{z} + t\epsilon, \quad \mathbf{v}_t^* = \epsilon - \mathbf{z}. \quad (1)$$

Given a text condition $\mathbf{y}$, the DiT $\mathbf{v}_\theta$ predicts this velocity by minimizing

$$\mathcal{L}_{\text{FM}} = \mathbb{E}_{\mathbf{z}, t, \epsilon} \left[\|\mathbf{v}_\theta(\mathbf{z}_t, t \mid \mathbf{y}) - \mathbf{v}_t^*\|_2^2 \right]. \quad (2)$$

Chunk-wise autoregressive video generation. To extend generation beyond a fixed clip, autoregressive methods divide a long video into fixed-length chunks and generate them sequentially. At a generic rollout step, we omit the rollout index from all quantities for clarity. Let $\mathbf{z}$ denote the clean latent of the current chunk, $\mathbf{z}_t$ its state at diffusion time t , $\mathbf{z}^h$ the accumulated clean history frames, and $\mathbf{z}^r$ the fixed-length recent history frames retained from $\mathbf{z}^h$. Given the text condition $\mathbf{y}$, a standard chunk-wise autoregressive model evolves the current latent through the conditional flow

$$\frac{d\mathbf{z}_t}{dt} = \mathbf{v}_\theta(\mathbf{z}_t, t \mid \mathbf{z}^r, \mathbf{y}). \quad (3)$$

At each denoising step, the DiT processes the concatenated latent sequence $[\mathbf{z}^r; \mathbf{z}_t]$, where $[\cdot; \cdot]$ denotes concatenation along the token sequence. After denoising, $\mathbf{z}$ is appended to $\mathbf{z}^h$, and the sliding window of $\mathbf{z}^r$ is updated with $\mathbf{z}$.

3.2. Base Architecture

To enable long-horizon memory and user interaction, we extend the standard autoregressive flow in Eq. (3) by conditioning each chunk jointly on a memory $\mathbf{M}$ derived from the accumulated history $\mathbf{z}^h$ and a target camera trajectory $\mathbf{C}$:

$$\frac{d\mathbf{z}_t}{dt} = \mathbf{v}_\theta(\mathbf{z}_t, t \mid \mathbf{M}, \mathbf{z}^r, \mathbf{C}, \mathbf{y}). \quad (4)$$

The overview of our pipeline is shown in Figure 2. The video DiT generates the first chunk conditioned on camera poses alone, as no history is yet available. As history accumulates, we learn a memory encoder to map history latent frames into a compact 3D-aware representation, which is read out as a fixed-size memory $\mathbf{M}$ to condition subsequent generation.

Memory conditioning. At each denoising step, the DiT processes $[\mathbf{M}; \mathbf{z}^r; \mathbf{z}_t]$ as a single latent sequence. This adds a dedicated memory stream to the recent history while preserving chunk-level causality. Unlike prior context-based memory [49, 59, 63, 70] that instantiate $\mathbf{M}$ as a fixed-length context retrieved from the accumulated history $\mathbf{z}^h$, we model $\mathbf{M}$ by mapping $\mathbf{z}^h$ into a compact 3D-aware implicit memory representation.

Camera conditioning. The target trajectory $\mathbf{C}$ specifies a camera-to-world pose and camera intrinsics for each frame. Following PROPE [32], we encode relative camera geometry as a positional transformation within self-attention.

We implement this conditioning using the parallel camera-attention branch of UCPE [78], which adopts independent query, key, and value projections and adds its output to the original self-attention through a zero-initialized projection. The camera branch is applied only to the noisy part $\mathbf{z}_t$ of the concatenated sequence; the memory $\mathbf{M}$ and recent context $\mathbf{z}^r$ are processed without camera injection.

3.3. Memory Encoder

Memory writing. As shown in Figure 2, after generating the first chunk, the memory encoder Φ writes the accumulated history latents $\mathbf{z}^h$ and their corresponding camera parameters $\mathbf{C}^h$ into an implicit 3D-aware representation $\mathbf{R}$:

$$\mathbf{R} = \Phi(\mathbf{z}^h, \mathbf{C}^h) \in \mathbb{R}^{|\mathbf{z}^h|L \times d}, \quad (5)$$

where $|\mathbf{z}^h|$ denotes the number of history latent frames. The encoder produces L tokens of dimension d per history latent frame.

We initialize the encoder architecture and weights from the LagerNVS encoder [50], discarding its shallow image-processing layers and adding a new patch embedding layer to map each latent frame directly into its representation space. The history camera poses are expressed relative to the latest latent frame in $\mathbf{z}^h$ and injected as camera tokens during encoding. The resulting representation tokens $\mathbf{R}$ aggregate geometry and appearance information across the input history without materializing an explicit 3D reconstruction.

Since the length of $\mathbf{R}$ grows linearly with the number of input history frames $|\mathbf{z}^h|$, to bound the cost of the encoder, we restrict its input to k history latent frames. At inference, we retain the latest latent frame in $\mathbf{z}^h$ and greedily select $k - 1$ complementary frames whose joint FoV maximizes coverage of the target region along the upcoming camera trajectory. We denote the selected history latents and their camera parameters by $\mathbf{z}^s$ and $\mathbf{C}^s$, respectively. Prior context-based memory methods [49, 59, 63, 70] retrieve only a few history frames by ranking their pairwise FoV similarity to target poses [59, 70], resulting in limited coverage and sensitivity to individual selections. In contrast, our memory encoder accommodates more history frames under a comparable budget, while max-coverage history retrieval yields broader coverage and greater robustness.

Memory readout. The written representation $\mathbf{R}$ should then be read out as a fixed-size memory $\mathbf{M}$ that conditions the DiT. We compare two readout mechanisms under a fixed DiT token budget.

The first is pose-free readout, in which a learned readout module maps the complete representation into a fixed set of memory tokens compatible with the DiT input:

$$\mathbf{M} = \text{Readout}(\Phi(\mathbf{z}^s, \mathbf{C}^s)). \quad (6)$$

This readout manner is independent of the upcoming camera trajectory, leaving the DiT attention to identify information relevant to the current generation.

The second is pose-guided readout, which queries the representation using a fixed-size of query poses $\mathbf{C}^q \subset \mathbf{C}$ sampled from the upcoming target camera trajectory:

$$\mathbf{M} = \text{Readout}(\Phi(\mathbf{z}^s, \mathbf{C}^s), \mathbf{C}^q). \quad (7)$$

In pose-guided readout, we initialize the readout module with the shallow decoder layers of LagerNVS [50] and add projection layers to map the output tokens into the DiT token space.

Empirically, we find that pose-guided readout outperforms pose-free readout and yields more accurate camera control. We attribute this gain to a more effective allocation of the fixed memory budget to target-relevant information. We therefore adopt pose-guided readout, with ablations presented in Sec. 4.5.

3.4. Base Model Training

Dataset curation. Our training data combine the OpenSora-Plan (OSP) dataset [34], DL3DV [36] dataset, and synthetic videos from MIND [66] dataset, covering diverse indoor and outdoor scenes and object motions. We use Depth Anything 3 [35] to obtain metric-scale camera pose annotations across all data sources and Qwen2.5-VL [4] to generate video captions. Using these captions and the estimated camera trajectories, we further curate a subset of the OSP dataset in which the camera follows moving subjects, helping the model learn coordinated camera and subject motion.

Training details. We initialize the video DiT from Helios-base [77] and train our base model in four stages. The original Helios-base inference window contains a 9-frame noise chunk and a FramePack-style clean history [79] comprising a compressed 16-frame segment, a 2-frame segment, the latest latent frame, and an attention-sink frame. We remove its compressed 16-frame segment and prepend memory tokens equivalent in number to the tokens of 4 uncompressed history frames.

In the first stage, we fine-tune Helios-base to adapt to our modified inference window. For each sampled video chunk, we randomly select 4 history frames from the preceding 4 chunks to populate the memory slots. We train on 760,000 videos from the OSP dataset for 5,000 iterations using 32 GPUs with a global batch size of 32.

In the second stage, we introduce camera control by training the UCPE-based camera conditioning branch while keeping the video DiT backbone frozen. We use 40,000 videos from the filtered OSP subset and 6,000 videos from DL3DV, training on 32 GPUs with a global batch size of 128.

In the third stage, we adapt our memory encoder to process VAE latents. We initialize it from the LagerNVS encoder and replace its shallow DINO layers with a latent patch embedding layer, as described in Sec. 3.3. The encoder takes a fixed number of 9 latent frames as input. We warm up the memory encoder on DL3DV and the filtered OSP subset for 5,000 iterations using 16 GPUs with a global batch size of 16.

In the final stage, we introduce a memory readout module that outputs a fixed number of memory tokens matching the token count of 4 full frames. We jointly train this module with the memory encoder, video DiT, and camera conditioning branch to co-adapt the learned memory representation and the DiT token space. We first train on DL3DV and the filtered OSP subset for 8,000 iterations using 32 GPUs with a global batch size of 32, then incorporate synthetic videos from MIND for 1,000 additional iterations to improve dynamic subject modeling.

3.5. Distillation for Real-time Interaction

Distill, low noise model and high noise model, 5+1 steps

4. Experiments

4.1. Experimental Setup

Benchmark. We curate a benchmark to evaluate memory ability, camera-control accuracy, and visual quality in long-horizon video world models. The benchmark contains 145 images from HappyOyster [14], Project Genie [5], web sources, and images generated by GPT-Image2, covering 83 dynamic object-centric scenes and 62 static scenes. Each image and its text description are paired with 5 metric camera trajectories, yielding 725 videos per method. The trajectories span 528–1,648 frames and include closed-loop revisits to assess whether previously observed content is preserved over long horizons.

Comparison methods. We compare WorldCrafter with 8 recent camera-controllable video world models: DreamX-World [12], EVOKE [69], HY-WorldPlay [49], Lyra 2.0 [48], Echo-WM [81], LingBot-World 2 [13], Matrix-Game 3.5 [46], and SANA-WM [83].

These baselines span different memory representations and camera-conditioning mechanisms. HY-WorldPlay and DreamX-World retrieve history context based on camera similarity and use PROPE [32] for camera control. SANA-WM combines Gated DeltaNet memory with UCPE [78] and Plücker-ray conditioning, whereas Echo-WM employs a UCPE-based camera branch and sliding-window memory. EVOKE and Lyra 2.0 construct spatial memory using depth estimated by Depth Anything 3 [35] and condition generation through geometric warping. Matrix-Game 3.5 combines geometric patch memory with Warped PROPE, using VGGT- Ω [55] and Depth Anything 3 for metric-scale ge-

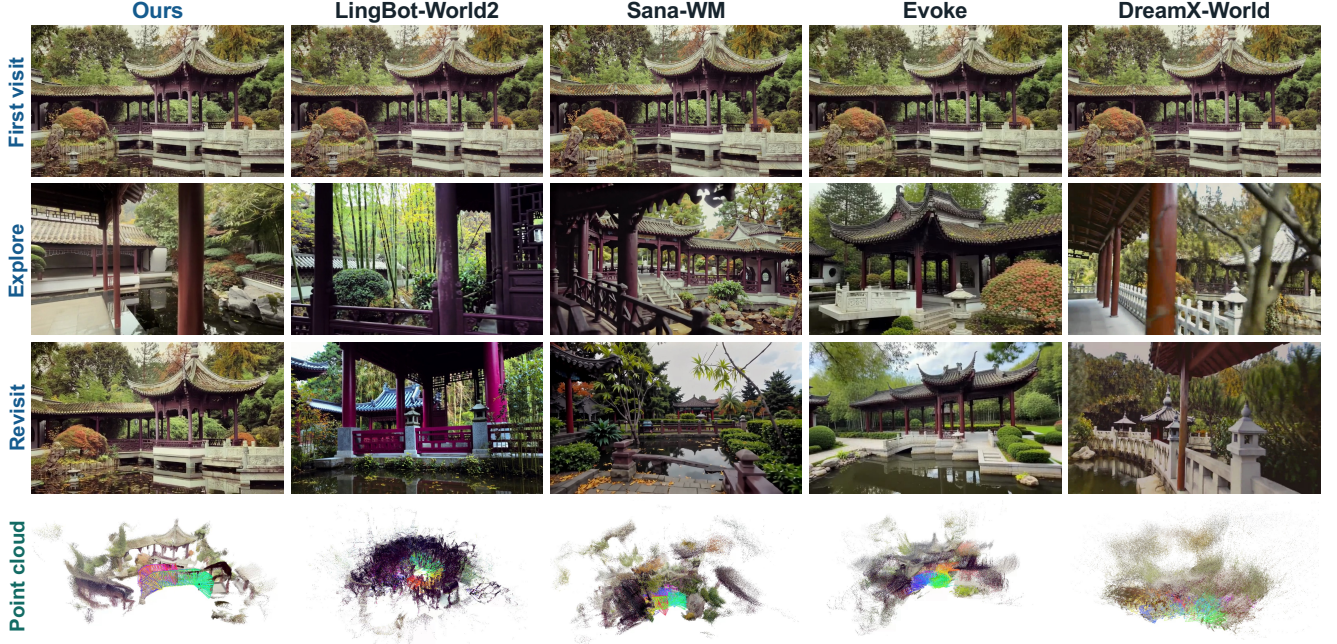


Figure 3. **Qualitative long-horizon revisit comparison on static scene.** Rows show the first observation, an intermediate view during exploration, the matched revisit, and the point cloud reconstruction of the generated video, respectively.

Table 1. **Long-horizon revisit consistency.** Results are averaged over 725 generated videos using matched first-visit and revisit frames. Dark and light blue mark the best and second-best results, respectively.

Method	MEt3R↓	LPIPS↓	PSNR↑	SSIM↑
DreamX-World [12]	0.548	0.627	12.898	0.243
EVOKE [69]	0.414	0.565	12.332	0.290
HY-WorldPlay [49]	0.394	0.515	12.983	0.252
Lyra 2.0 [48]	0.334	0.487	14.050	0.390
Echo-WM [81]	0.449	0.582	12.592	0.239
LingBot-World 2 [13]	0.492	0.633	10.449	0.219
Matrix-Game 3.5 [46]	0.405	0.549	12.976	0.224
SANA-WM [83]	0.397	0.553	13.142	0.246
WorldCrafter	0.166	0.255	18.016	0.517

Table 2. **Camera-control accuracy.** Results are averaged over 725 generated videos using Sim(3)-aligned estimated and target trajectories. Dark and light blue mark the best and second-best results, respectively.

Method	RotErr↓	TransErr↓	CamMC↓
DreamX-World [12]	54.116	2.759	3.138
EVOKE [69]	26.042	2.042	2.199
HY-WorldPlay [49]	34.051	2.146	2.359
Lyra 2.0 [48]	16.145	1.538	1.624
Echo-WM [81]	21.455	2.072	2.189
LingBot-World 2 [13]	30.615	2.004	2.192
Matrix-Game 3.5 [46]	19.881	1.920	2.028
SANA-WM [83]	23.531	1.740	1.887
WorldCrafter	13.536	1.475	1.546

ometry annotation.

We evaluate the full-step base model with a refiner for SANA-WM, the full-step base models for Lyra 2.0 and HY-WorldPlay, and the distilled models for the remaining base-lines. All methods receive identical initial images, text descriptions, and target trajectories. Before evaluation, we resize the generated videos to 640×384 to ensure a common evaluation resolution.

4.2. Memory Evaluation

Following the protocol in [51], we evaluate memory ability by comparing frames generated upon revisiting a location

with the corresponding frames from the initial visit. We report MEt3R [2], LPIPS [80], PSNR, and SSIM [56] to assess consistency between the paired observations.

As shown in Table 1, WorldCrafter achieves the best results on all 4 metrics. Compared with Lyra 2.0, WorldCrafter reduces LPIPS from 0.487 to 0.255 and increases PSNR from 14.050 to 18.016 dB. The agreement across these metrics indicates that revisited views more faithfully recover the appearance and structure of earlier observations, supporting the effectiveness of the learned memory representation in long-horizon closed-loop exploration. Figures 3 and 4 compare first-visit and revisit frames across

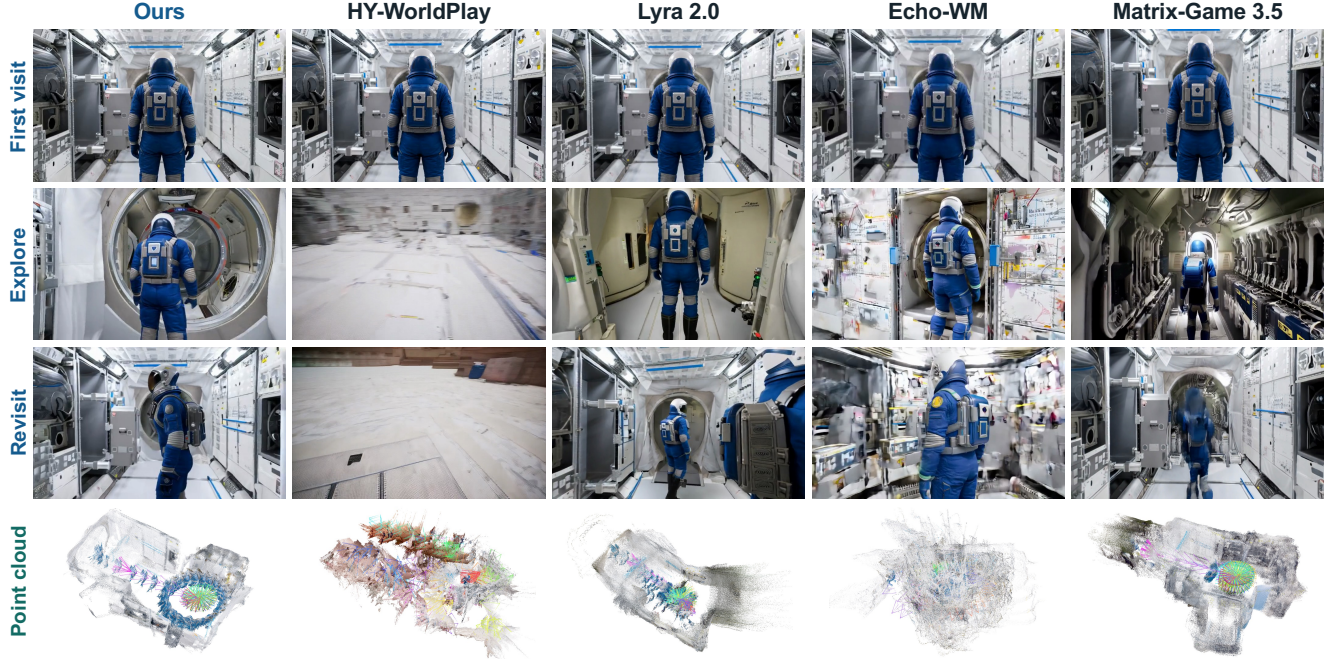


Figure 4. **Qualitative long-horizon revisit comparison on dynamic scene.** Rows show the first observation, an intermediate view during exploration, the matched revisit, and the point cloud reconstruction of the generated video, respectively.

Table 3. **Visual quality on VBench.** Scores are computed in custom-input mode over 725 generated videos. SC, BC, TF, MS, AQ, IQ, DD, and OC denote subject consistency, background consistency, temporal flickering, motion smoothness, aesthetic quality, imaging quality, dynamic degree, and overall consistency, respectively. The Overall column reports the aggregate VBench score. Dark and light blue mark the best and second-best results, respectively.

Method	SC	BC	TF	MS	AQ	IQ	DD	OC	Overall
DreamX-World [12]	75.994	87.601	95.327	98.100	57.593	60.412	99.806	24.583	77.652
EVOKE [69]	80.522	88.994	95.872	98.481	61.579	71.249	99.029	25.015	81.406
HY-WorldPlay [49]	75.407	86.629	95.553	97.626	53.454	61.700	97.087	21.182	76.539
Lyra 2.0 [48]	77.277	87.935	95.167	98.652	57.832	65.149	99.806	21.484	78.943
Echo-WM [81]	81.671	90.206	95.771	96.751	61.818	66.564	98.835	26.082	80.211
LingBot-World 2 [13]	75.584	87.725	94.206	96.657	60.815	68.776	100.000	25.605	78.174
Matrix-Game 3.5 [46]	70.370	83.626	94.395	98.087	58.615	69.374	100.000	18.813	76.967
SANA-WM [83]	79.413	89.625	95.768	98.608	63.467	66.080	98.835	25.234	80.841
WorldCrafter	82.695	90.589	96.217	98.787	61.432	69.058	96.893	26.745	81.910

methods on a 103-second closed-loop trajectory. WorldCrafter retains the pavilion’s structure and surrounding vegetation, whereas several baselines exhibit changes in scene layout or appearance.

4.3. Camera Control Evaluation

We sample the generated videos at a stride of 4 frames and recover camera trajectories using VGGT- Ω [55]. Following SANA-WM [83], each trajectory is normalized relative to its first pose and aligned using Umeyama Sim(3) alignment [52]. Evaluation metrics include rotation error (RotErr), camera-center translation error (TransErr),

and pose-matrix discrepancy (CamMC), with lower values indicating more accurate camera control. WorldCrafter achieves the lowest error on all 3 metrics in Table 2, indicating more accurate adherence to the input camera trajectories.

4.4. Visual Quality Evaluation

Following SANA-WM [83], we evaluate video generation quality using VBench [22] in custom-input mode. We report Subject Consistency (SC), Background Consistency (BC), Temporal Flickering (TF), Motion Smoothness (MS), Aesthetic Quality (AQ), Imaging Quality (IQ), Dynamic

Table 4. **Ablation of the memory design.** Each variant changes one design choice while keeping the training and inference settings and the DiT memory-token budget fixed. Arrows indicate the preferred direction; the full-model row is shaded.

Method	Memory evaluation				Camera evaluation		
	MEt3R↓	LPIPS↓	PSNR↑	SSIM↑	RotErr↓	TransErr↓	CamMC↓
Context memory	0.382	0.497	13.907	0.315	26.522	2.083	2.150
Frozen memory encoder	0.227	0.305	16.873	0.472	15.428	1.793	1.886
Pose-free memory readout	0.251	0.333	16.486	0.467	18.307	1.701	1.828
Similarity-based history retrieval	0.213	0.296	17.125	0.485	14.657	1.579	1.664
Ours	0.166	0.255	18.016	0.517	13.536	1.475	1.546

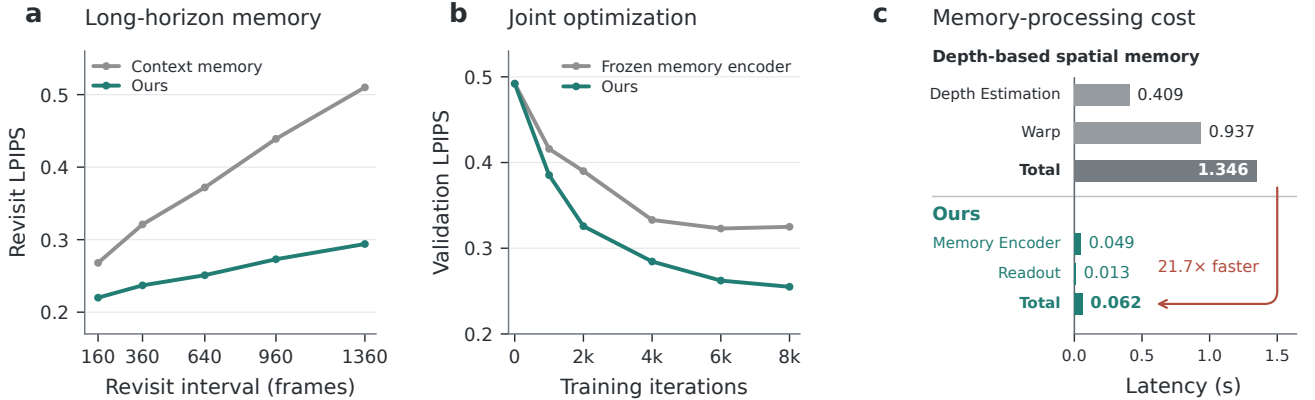


Figure 5. **Analysis of memory retention, optimization, and processing cost.** (a) Revisit LPIPS as a function of the temporal interval between matched first-visit and revisit frames for context memory and WorldCrafter. (b) Validation LPIPS at matched training iterations for joint and frozen memory-encoder training. (c) Per-chunk memory-processing latency at 640×384 resolution for depth-based spatial memory and WorldCrafter. The latency setting uses a 9-latent-frame generation chunk and 4 history frames for spatial warping; *Total* sums the displayed components. Video denoising and VAE decoding are shared and therefore excluded. Lower is better in all panels.

Degree (DD), and Overall Consistency (OC). As shown in Table 3, WorldCrafter achieves the highest overall score of 81.910, ranking first in 5 of the 8 dimensions. These gains primarily reflect stronger scene consistency and temporal coherence across the generated videos, complementing the results of the memory evaluation.

4.5. Ablation Study

All ablations follow the memory and camera-control evaluation protocols in Secs. 4.2 and 4.3. Table 4 compares the full model with 4 variants under the same training and inference settings.

Implicit 3D-aware memory versus context memory. The *context memory* variant replaces the memory tokens produced by the memory encoder and memory readout module with 4 retrieved history latent frames. It is initialized from the second-stage model in Sec. 3.4 and further trained for the same number of iterations as the full model. Table 4

shows that this replacement degrades both revisit consistency and camera control. To examine how memory retention varies over time, Figure 5(a) plots mean revisit LPIPS against the frame interval between the initial observation and its revisit. WorldCrafter exhibits a slower increase in revisit error, with a widening advantage over context memory at longer intervals.

Joint optimization of the memory encoder and video generator. The *frozen memory encoder* variant fixes the encoder during the final training stage, while keeping the memory readout module, video DiT, and camera conditioning branch trainable. This variant yields weaker memory and camera-control performance in Table 4. Figure 5(b) tracks validation performance at matched training iterations: joint optimization reaches lower revisit error earlier and maintains this advantage. This supports co-adapting the memory representation with the video generator instead of learning to consume a fixed representation space.

Memory readout. The *pose-free memory readout* variant maps the encoded representation to memory tokens without target-pose queries, retaining the same history inputs and output token budget. Table 4 shows that pose-guided readout improves both memory and camera control ability.

History retrieval. To isolate the effect of history retrieval, the *similarity-based history retrieval* variant replaces max-coverage retrieval with pairwise FoV similarity ranking. Both variants retain the latest latent frame and select 8 additional history frames as the memory encoder input, with the same DiT memory-token budget. Table 4 shows improved revisit consistency and camera control with max-coverage retrieval. This comparison favors selecting complementary views that jointly cover the target region over independently ranking views by similarity, without increasing the number of history inputs.

Memory efficiency. Depth-based spatial memory methods, including Lyra 2.0 [48], Matrix-Game 3.5 [46], and EVOKE [69], rely on estimated geometry to reuse history observations. We compare the additional cost of depth estimation and warping with that of WorldCrafter’s memory encoder and readout. Figure 5(c) reports component latencies at 640×384 resolution, with a 9 latent frames generation chunk and 4 history frames for spatial warping. Shared VAE decoding for RGB output and video denoising are excluded. For spatial memory, the depth estimation and alignment process takes 0.409 s using Depth Anything 3 [35], while the batched warping takes 0.937 s, totaling 1.346 s per chunk. By operating directly on history latents, WorldCrafter requires 0.049 s for memory encoding and 0.013 s for readout, totaling 0.062 s. This reduces the summed memory-processing cost by a factor of $21.7\times$, without requiring explicit depth estimation or warping.

5. Discussion

We present WorldCrafter, a camera-controllable autoregressive video world model with implicit 3D-aware memory. A learned memory encoder aggregates history latent frames into a compact representation, which is read out as tokens compatible with the video DiT. Experiments demonstrate improved revisit consistency and camera-control accuracy over the evaluated baselines.

Several limitations remain. Consistency can still break down along particularly complex or extended trajectories. In addition, re-encoding history at every chunk incurs extra latency. A promising direction is a streaming memory encoder that incrementally incorporates each newly generated chunk into the memory state, reducing repeated computation over history.

References

[1] AlayaWorld Team, Kaipeng Zhang, Chuanhao Li, Yifan

Zhan, Yongtao Ge, Yuanyang Yin, Jiaming Tan, Kang He, Liaoyuan Fan, Ruicong Liu, Xiaojie Xu, Xuangeng Chu, Zhen Li, Zhengyuan Lin, Zhixiang Wang, Zian Meng, and Zihui Gao. AlayaWorld: Long-horizon and playable video world generation. *arXiv preprint arXiv:2607.06291*, 2026. 2

[2] Mohammad Asim, Christopher Wewer, Thomas Wimmer, Bernt Schiele, and Jan Eric Lenssen. MET3R: Measuring multi-view consistency in generated images. In *CVPR*, pages 6034–6044, 2025. 6

[3] Jianhong Bai, Menghan Xia, Xiao Fu, Xintao Wang, Lianrui Mu, Jinwen Cao, Zuozhu Liu, Haoji Hu, Xiang Bai, Pengfei Wan, and Di Zhang. ReCamMaster: Camera-controlled generative rendering from a single video. In *ICCV*, pages 14834–14844, 2025. 2

[4] Shuai Bai, Keqin Chen, Xuejing Liu, Jialin Wang, Wenbin Ge, Sibao Song, Kai Dang, Peng Wang, Shijie Wang, Jun Tang, Humen Zhong, Yuanzhi Zhu, Mingkun Yang, Zhao-hai Li, Jianqiang Wan, Pengfei Wang, Wei Ding, Zheren Fu, Yiheng Xu, Jiabo Ye, Xi Zhang, Tianbao Xie, Zesen Cheng, Hang Zhang, Zhibo Yang, Haiyang Xu, and Junyang Lin. Qwen2.5-VL technical report. *arXiv preprint arXiv:2502.13923*, 2025. 5

[5] Philip J. Ball, Jakob Bauer, Frank Belletti, et al. Genie 3: A new frontier for world models, 2025. 2, 5

[6] Shengqu Cai, Ceyuan Yang, Lvmin Zhang, Yuwei Guo, Junfei Xiao, Ziyang Yang, Yinghao Xu, Zhenheng Yang, Alan Yuille, Leonidas Guibas, Maneesh Agrawala, Lu Jiang, and Gordon Wetzstein. Mixture of contexts for long video generation. *arXiv preprint arXiv:2508.21058*, 2025. 2, 3

[7] Boyuan Chen, Diego Martí Monsó, Yilun Du, Max Simchowitz, Russ Tedrake, and Vincent Sitzmann. Diffusion forcing: Next-token prediction meets full-sequence diffusion. In *NeurIPS*, 2024. 2

[8] Kaijin Chen, Dingkan Liang, Xin Zhou, Yikang Ding, Xiaoliang Liu, Pengfei Wan, and Xiang Bai. Out of sight but not out of mind: Hybrid memory for dynamic video world models. *arXiv preprint arXiv:2603.25716*, 2026. 2, 3

[9] Yukang Chen, Luozhou Wang, Wei Huang, Shuai Yang, Bohan Zhang, Yicheng Xiao, Ruihang Chu, Weian Mao, Qixin Hu, Shaoteng Liu, Yuyang Zhao, Huizi Mao, Ying-Cong Chen, Enze Xie, Xiaojuan Qi, and Song Han. Longlive-2.0: An NVFP4 parallel infrastructure for long video generation. *arXiv preprint arXiv:2605.18739*, 2026. 2

[10] Zhifei Chen, Luozhou Wang, Guibao Shen, Dongyu Yan, Shuai Yang, Tianshuo Xu, Yihua Du, Wei Wang, Tianyi Gui, Lianghua Huang, and Yingcong Chen. ReWorld: An interactive world model with long-horizon memory. *arXiv preprint arXiv:2608.23565*, 2026. 2

[11] Decart and Etched. Oasis: A universe in a transformer. Project page: <https://oasis-model.github.io/>, 2024. 2

[12] DreamX Team, Yancheng Bai, Rui Chen, Xiangxiang Chu, Rujing Dang, Hao Dou, Bingjie Gao, Qiwen Gu, Siyu Hong, Jiachen Lei, Geng Li, Jifan Li, Ruimin Lin, Qingfeng Shi, Bingze Song, Lei Sun, Jing Tang, Ruitian Tian, Jun Wang, Jiahong Wu, Pengfei Zhang, Shen Zhang, and Jiashu Zhu. DreamX-World 1.0: A general-purpose interactive world model. *arXiv preprint arXiv:2606.16993*, 2026. 2, 5, 6, 7

- [13] Zelin Gao, Qiuyu Wang, Jiapeng Zhu, Jingye Chen, Zichen Liu, Qingyan Bai, Jiahao Wang, Yufeng Yuan, Hanlin Wang, Yichong Lu, Ka Leong Cheng, Haojie Zhang, Jian Gao, Tianrui Feng, Yuzheng Liu, Yao Yao, Yinghao Xu, Xing Zhu, Yujun Shen, and Hao Ouyang. Infinite worlds with versatile interactions. *arXiv preprint arXiv:2607.07534*, 2026. 5, 6, 7
- [14] Happy Oyster. Happy oyster: Real-time world model for interactive creation, 2026. 2, 5
- [15] Hao He, Yinghao Xu, Yuwei Guo, Gordon Wetzstein, Bo Dai, Hongsheng Li, and Ceyuan Yang. Cameractrl: Enabling camera control for text-to-video generation. *arXiv preprint arXiv:2404.02101*, 2024. 2
- [16] Roberto Henschel, Levon Khachatryan, Hayk Poghosyan, Daniil Hayrapetyan, Vahram Tadevosyan, Zhangyang Wang, Shant Navasardyan, and Humphrey Shi. StreamingT2V: Consistent, dynamic, and extendable long video generation from text. In *CVPR*, pages 2568–2577, 2025. 2
- [17] Yicong Hong, Yiqun Mei, Chongjian Ge, Yiran Xu, Yang Zhou, Sai Bi, Yannick Hold-Geoffroy, Mike Roberts, Matthew Fisher, Eli Shechtman, Kalyan Sunkavalli, Feng Liu, Zhengqi Li, and Hao Tan. Relic: Interactive video world model with long-horizon memory. *arXiv preprint arXiv:2512.04040*, 2025. 2, 3
- [18] Junchao Huang, Xinting Hu, Boyao Han, Shaoshuai Shi, Zhuotao Tian, Tianyu He, and Li Jiang. Memory forcing: Spatio-temporal memory for consistent scene generation on minecraft. *arXiv preprint arXiv:2510.03198*, 2025. 3
- [19] Kaiyi Huang, Yukun Huang, Yu Li, Jianhong Bai, Xintao Wang, Zinan Lin, Xuefei Ning, Jiwen Yu, Pengfei Wan, Yu Wang, and Xihui Liu. Cinescene: Implicit 3d as effective scene representation for cinematic video generation. *arXiv preprint arXiv:2602.06959*, 2026. 2, 3
- [20] Xun Huang. Towards video world models, 2025. 2
- [21] Xun Huang, Zhengqi Li, Guande He, Mingyuan Zhou, and Eli Shechtman. Self forcing: Bridging the train-test gap in autoregressive video diffusion. *arXiv preprint arXiv:2506.08009*, 2025. 2
- [22] Ziqi Huang, Yinan He, Jiashuo Yu, Fan Zhang, Chenyang Si, Yuming Jiang, Yuanhan Zhang, Tianxing Wu, Qingyang Jin, Nattapol Chanpaisit, Yaohui Wang, Xinyuan Chen, Limin Wang, Dahua Lin, Yu Qiao, and Ziwei Liu. VBench: Comprehensive benchmark suite for video generative models. In *CVPR*, pages 21807–21818, 2024. 7
- [23] Fan Jiang, Zhaoxu Sun, Mengchao Wang, Ziyu Zhu, Chiyu Wang, Yunpeng Zhang, Wenlin Liu, Yun Wang, Xue Zheng, Rui Sun, et al. ABot-World-0: Infinite interactive world rollout on a single desktop GPU. *arXiv preprint arXiv:2607.19191*, 2026. 2
- [24] Hanwen Jiang, Hao Tan, Peng Wang, Haian Jin, Yue Zhao, Sai Bi, Kai Zhang, Fujun Luan, Kalyan Sunkavalli, Qixing Huang, and Georgios Pavlakos. RayZer: A self-supervised large view synthesis model. In *ICCV*, 2025. 2
- [25] Haian Jin, Hanwen Jiang, Hao Tan, Kai Zhang, Sai Bi, Tianyuan Zhang, Fujun Luan, Noah Snively, and Zexiang Xu. LVSM: A large view synthesis model with minimal 3d inductive bias. In *ICLR*, 2025.
- [26] Evan Kim, Hyunwoo Ryu, Thomas W. Mitchel, and Vincent Sitzmann. Scaling view synthesis transformers. In *CVPR*, 2026. 2
- [27] Jihwan Kim, Junoh Kang, Jinyoung Choi, and Bohyung Han. FIFO-Diffusion: Generating infinite videos from text without training. In *NeurIPS*, 2024. 2
- [28] Diederik P. Kingma and Max Welling. Auto-encoding variational bayes. In *ICLR*, 2014. 3
- [29] Jiaqi Li, Junshu Tang, Zhiyong Xu, Longhuang Wu, Yuan Zhou, Shuai Shao, Tianbao Yu, Zhiguo Cao, and Qinglin Lu. Hunyuan-gamecraft: High-dynamic interactive game video generation with hybrid history condition. *arXiv preprint arXiv:2506.17201*, 2025. 2
- [30] Jia Li, Han Yan, Yihang Chen, Siqi Li, Xibin Song, Yifu Wang, Jianfei Cai, Tien-Tsin Wong, and Pan Ji. I3DM: Implicit 3d-aware memory retrieval and injection for consistent video scene generation. *arXiv preprint arXiv:2603.23413*, 2026. 3
- [31] Runjia Li, Philip Torr, Andrea Vedaldi, and Tomas Jakab. Vmem: Consistent interactive video scene generation with surfel-indexed view memory. In *ICCV*, pages 25690–25699, 2025. 2, 3
- [32] Ruilong Li, Brent Yi, Junchen Liu, Hang Gao, Yi Ma, and Angjoo Kanazawa. Cameras as relative positional encoding. In *NeurIPS*, 2025. 2, 4, 5
- [33] Zhiqi Li, Chengrui Dong, Zhenhua Du, Hangning Zhou, Cong Qiu, Hailong Qin, Mu Yang, Dongxu Wei, and Peidong Liu. Walking in the implicit: Interactive world exploration via neural scene representation. *arXiv preprint arXiv:2606.30045*, 2026. 3
- [34] Bin Lin, Yunyang Ge, Xinhua Cheng, Zongjian Li, Bin Zhu, Shaodong Wang, Xianyi He, Yang Ye, Shenghai Yuan, Lihuan Chen, et al. Open-Sora Plan: Open-source large video generation model. *arXiv preprint arXiv:2412.00131*, 2024. 5
- [35] Haotong Lin, Sili Chen, Junhao Liew, Donny Y. Chen, Zhenyu Li, Guang Shi, Jiashi Feng, and Bingyi Kang. Depth anything 3: Recovering the visual space from any views. *arXiv preprint arXiv:2511.10647*, 2025. 5, 9
- [36] Lu Ling, Yichen Sheng, Zhi Tu, Wentian Zhao, Cheng Xin, Kun Wan, Lantao Yu, Qianyu Guo, Zixun Yu, Yawen Lu, et al. DL3DV-10K: A large-scale scene dataset for deep learning-based 3d vision. In *CVPR*, pages 22160–22169, 2024. 5
- [37] Yaron Lipman, Ricky T. Q. Chen, Heli Ben-Hamu, Maximilian Nickel, and Matthew Le. Flow matching for generative modeling. In *ICLR*, 2023. 3
- [38] Kunhao Liu, Wenbo Hu, Jiale Xu, Ying Shan, and Shijian Lu. Rolling forcing: Autoregressive long video diffusion in real time. *arXiv preprint arXiv:2509.25161*, 2025. 2
- [39] Xiaofeng Mao, Shaoheng Lin, Zhen Li, Chuanhao Li, Wenshuo Peng, Tong He, Jiangmiao Pang, Mingmin Chi, Yu Qiao, and Kaipeng Zhang. Yume: An interactive world generation model. *arXiv preprint arXiv:2507.17744*, 2025. 2
- [40] NVIDIA, Aditi, Niket Agarwal, Arslan Ali, Jon Allen, Martin Antolini, Adeline Aubame, Alisson Azzolini, Junjie Bai, Maciej Bala, Yogesh Balaji, Josh Bapst, et al. Cosmos 3:

- Omnimodal world models for physical AI. *arXiv preprint arXiv:2606.02800*, 2026. 2
- [41] Yuta Oshima, Yusuke Iwasawa, Masahiro Suzuki, Yutaka Matsuo, and Hiroki Furuta. Worldpack: Compressed memory improves spatial consistency in video world modeling. *arXiv preprint arXiv:2512.02473*, 2025. 3
- [42] Jack Parker-Holder, Philip Ball, Jake Bruce, Vibhavari Dasagi, Kristian Holsheimer, Christos Kaplanis, Alexandre Moufarek, Guy Scully, Jeremy Shar, Jimmy Shi, Stephen Spencer, Jessica Yung, Michael Dennis, Sultan Kenjeyev, Shangbang Long, Vlad Mnihi, Harris Chan, Maxime Gazeau, Bonnie Li, Fabio Pardo, Luyu Wang, Lei Zhang, Frederic Besse, Tim Harley, Anna Mitenkova, Jane Wang, Jeff Clune, Demis Hassabis, Raia Hadsell, Adrian Bolton, Satinder Singh, and Tim Rocktäschel. Genie 2: A large-scale foundation world model. Google DeepMind project page, 2024. 2
- [43] William Peebles and Saining Xie. Scalable diffusion models with transformers. In *ICCV*, pages 4195–4205, 2023. 3
- [44] Zhan Peng, Jie Ma, Huiqiang Sun, Chong Gao, Zhijie Xue, Zhiyu Pan, Zhiguo Cao, Jun Liang, and Jing Li. Compression and retrieval: Implicit memory retrieval for video world models. *arXiv preprint arXiv:2606.23105*, 2026. 2, 3
- [45] Xuanchi Ren, Tianchang Shen, Jiahui Huang, Huan Ling, Yifan Lu, Merlin Nimier-David, Thomas Müller, Alexander Keller, Sanja Fidler, and Jun Gao. Gen3c: 3d-informed world-consistent video generation with precise camera control. In *CVPR*, pages 6121–6132, 2025. 2, 3
- [46] Riemann Dynamics. Matrix-Game 3.5: Enhancing real-time streaming interactive world models with patch memory. Technical report, 2026. 2, 5, 6, 7, 9
- [47] Nedko Savov, Naser Kazemi, Deheng Zhang, Danda Pani Paudel, Xi Wang, and Luc Van Gool. Statespacediffuser: Bringing long context to diffusion world models. *arXiv preprint arXiv:2505.22246*, 2025. 2, 3
- [48] Tianchang Shen, Sherwin Bahmani, Kai He, Sangeetha Grama Srinivasan, Tianshi Cao, Jiawei Ren, Ruilong Li, Zian Wang, Nicholas Sharp, Zan Gojcic, Sanja Fidler, Jiahui Huang, Huan Ling, Jun Gao, and Xuanchi Ren. Lyra 2.0: Explorable generative 3d worlds. *arXiv preprint arXiv:2604.13036*, 2026. 2, 5, 6, 7, 9
- [49] Wenqiang Sun, Haiyu Zhang, Haoyuan Wang, Junta Wu, Zehan Wang, Zhenwei Wang, Yunhong Wang, Jun Zhang, Tengfei Wang, and Chunchao Guo. Worldplay: Towards long-term geometric consistency for real-time interactive world modeling. *arXiv preprint arXiv:2512.14614*, 2025. 2, 3, 4, 5, 6, 7
- [50] Stanisław Szymanowicz, Minghao Chen, Jianyuan Wang, Christian Rupprecht, and Andrea Vedaldi. LagerNVS: Latent geometry for fully neural real-time novel view synthesis. In *CVPR*, pages 15443–15453, 2026. 2, 3, 4, 5
- [51] Team HunyuanWorld. Hy-world 1.5: A systematic framework for interactive world modeling with real-time latency and geometric consistency. *Technical report*, 2025. 2, 6
- [52] Shinji Umeyama. Least-squares estimation of transformation parameters between two point patterns. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 13(4):376–380, 1991. 7
- [53] Team Wan, Ang Wang, Baole Ai, Bin Wen, Chaojie Mao, Chen-Wei Xie, Di Chen, Feiwei Yu, Haiming Zhao, Jianxiao Yang, et al. Wan: Open and advanced large-scale video generative models. *arXiv preprint arXiv:2503.20314*, 2025. 3
- [54] Jianyuan Wang, Minghao Chen, Nikita Karaev, Andrea Vedaldi, Christian Rupprecht, and David Novotny. VGGT: Visual geometry grounded transformer. In *CVPR*, pages 5294–5306, 2025. 2, 3
- [55] Jianyuan Wang, Minghao Chen, Shangzhan Zhang, Nikita Karaev, Johannes Schöonberger, Patrick Labatut, Piotr Bojanowski, David Novotny, Andrea Vedaldi, and Christian Rupprecht. VGGT-Ω. In *CVPR*, 2026. 5, 7
- [56] Zhou Wang, Alan C. Bovik, Hamid R. Sheikh, and Eero P. Simoncelli. Image quality assessment: From error visibility to structural similarity. *IEEE Transactions on Image Processing*, 13(4):600–612, 2004. 6
- [57] Zhouxia Wang, Ziyang Yuan, Xintao Wang, Yaowei Li, Tianshui Chen, Menghan Xia, Ping Luo, and Ying Shan. Motionctrl: A unified and flexible motion controller for video generation. In *SIGGRAPH*, pages 1–11, 2024. 2
- [58] Zun Wang, Han Lin, Jaehong Yoon, Jaemin Cho, Yue Zhang, and Mohit Bansal. Anchorweave: World-consistent video generation with retrieved local spatial memories. *arXiv preprint arXiv:2602.14941*, 2026. 3
- [59] Zile Wang, Zexiang Liu, Jiaying Li, Kaichen Huang, Baixin Xu, Fei Kang, Mengyin An, Peiyu Wang, Biao Jiang, Yichen Wei, et al. Matrix-game 3.0: Real-time and streaming interactive world model with long-horizon memory. *arXiv preprint arXiv:2604.08995*, 2026. 2, 3, 4
- [60] Zhengxuan Wei, Xu Guo, Xinghui Li, Xunzhi Xiang, Min Wei, Yiran Zhu, Qiulin Wang, Xintao Wang, Pengfei Wan, Xiangwang Hou, and Qi Fan. Geometry-aware implicit memory for video world models. *arXiv preprint arXiv:2606.02436*, 2026. 2, 3
- [61] Ruiqi Wu, Xuanhua He, Meng Cheng, Tianyu Yang, Yong Zhang, Zhuoliang Kang, Xunliang Cai, Xiaoming Wei, Chunle Guo, Chongyi Li, and Ming-Ming Cheng. Infinite-world: Scaling interactive world models to 1000-frame horizons via pose-free hierarchical memory. *arXiv preprint arXiv:2602.02393*, 2026. 2, 3
- [62] Tong Wu, Shuai Yang, Ryan Po, Yinghao Xu, Ziwei Liu, Dahua Lin, and Gordon Wetzstein. Video world models with long-term spatial memory. *arXiv preprint arXiv:2506.05284*, 2025. 2, 3
- [63] Zeqi Xiao, Yushi Lan, Yifan Zhou, Wenqi Ouyang, Shuai Yang, Yanhong Zeng, and Xingang Pan. Worldmem: Long-term consistent world simulation with memory. *arXiv preprint arXiv:2504.12369*, 2025. 2, 3, 4
- [64] Tianxing Xu, Zixuan Wang, Guangyuan Wang, Li Hu, Zhongyi Zhang, Peng Zhang, Bang Zhang, and Song-Hai Zhang. Ucm: Unifying camera control and memory with time-aware positional encoding warping for world models. *arXiv preprint arXiv:2602.22960*, 2026. 3
- [65] Shuai Yang, Wei Huang, Ruihang Chu, Yicheng Xiao, Yuyang Zhao, Xianbang Wang, Muyang Li, Enze Xie, Yingcong Chen, Yao Lu, Song Han, and Yukang Chen. Longlive:

- Real-time interactive long video generation. In *ICLR*, 2026. 2
- [66] Yixuan Ye, Xuanyu Lu, Yuxin Jiang, Yuchao Gu, Rui Zhao, Qiwei Liang, Jiachun Pan, Fengda Zhang, Weijia Wu, and Alex Jinpeng Wang. MIND: Benchmarking memory consistency and action control in world models. *arXiv preprint arXiv:2602.08025*, 2026. 5
- [67] Jung Yi, Minjae Kim, Paul Hyunbin Cho, Wooseok Jang, Sangdoo Yun, and Seungryong Kim. WorldKV: Efficient world memory with world retrieval and compression. *arXiv preprint arXiv:2605.22718*, 2026. 2
- [68] Tianwei Yin, Qiang Zhang, Richard Zhang, William T. Freeman, Frédo Durand, Eli Shechtman, and Xun Huang. From slow bidirectional to fast autoregressive video diffusion models. In *CVPR*, 2025. 2
- [69] Yuanyang Yin, Gongxuan Wang, Yifan Zhan, Chuanhao Li, Kaipeng Zhang, and Feng Zhao. Alaya-EVOKE: From linear-scaling supervision to endless world. *arXiv preprint arXiv:2608.13546*, 2026. 2, 3, 5, 6, 7, 9
- [70] Jiwen Yu, Jianhong Bai, Yiran Qin, Quande Liu, Xintao Wang, Pengfei Wan, Di Zhang, and Xihui Liu. Context as memory: Scene-consistent interactive long video generation with memory retrieval. In *SIGGRAPH Asia*, pages 1–11, 2025. 2, 3, 4
- [71] Jiwen Yu, Yiran Qin, Xintao Wang, Pengfei Wan, Di Zhang, and Xihui Liu. Gamefactory: Creating new games with generative interactive videos. In *ICCV*, pages 11590–11599, 2025. 2
- [72] Jiwen Yu, Jianxiong Gao, Jianhong Bai, Yiran Qin, Kaiyi Huang, Quande Liu, Xintao Wang, Pengfei Wan, Kun Gai, and Xihui Liu. Memlearner: Learning to query context memory for video world models. *arXiv preprint arXiv:2606.31734*, 2026. 2, 3
- [73] Mark Yu, Wenbo Hu, Jinbo Xing, and Ying Shan. Trajectorycrafter: Redirecting camera trajectory for monocular videos via diffusion models. In *ICCV*, pages 100–111, 2025. 2
- [74] Wangbo Yu, Jinbo Xing, Li Yuan, Wenbo Hu, Xiaoyu Li, Zhipeng Huang, Xiangjun Gao, Tien-Tsin Wong, Ying Shan, and Yonghong Tian. Viewcrafter: Taming video diffusion models for high-fidelity novel view synthesis. *arXiv preprint arXiv:2409.02048*, 2024. 2, 3
- [75] Wei Yu, Runjia Qian, Yumeng Li, Liquan Wang, Songheng Yin, Dennis Anthony, Yang Ye, Yidi Li, Weiwei Wan, Animesh Garg, et al. Mosaicmem: Hybrid spatial memory for controllable video world models. *arXiv preprint arXiv:2603.17117*, 2026. 2, 3
- [76] Yifei Yu, Xiaoshan Wu, Xinting Hu, Tao Hu, Yangtian Sun, Xiaoyang Lyu, Bo Wang, Lin Ma, Yuewen Ma, Zhongrui Wang, et al. Videoss: Autoregressive long video generation with hybrid state-space memory. *arXiv preprint arXiv:2512.04519*, 2025. 2, 3
- [77] Shenghai Yuan, Yuanyang Yin, Zongjian Li, Xinwei Huang, Xiao Yang, and Li Yuan. Helios: Real real-time long video generation model. *arXiv preprint arXiv:2603.04379*, 2026. 5
- [78] Cheng Zhang, Boying Li, Meng Wei, Yan-Pei Cao, Camilo Cruz Gambardella, Dinh Phung, and Jianfei Cai. Unified camera positional encoding for controlled video generation. *arXiv preprint arXiv:2512.07237*, 2025. 2, 4, 5
- [79] Lvmin Zhang and Maneesh Agrawala. Packing input frame context in next-frame prediction models for video generation. *arXiv preprint arXiv:2504.12626*, 2025. 3, 5
- [80] Richard Zhang, Phillip Isola, Alexei A. Efros, Eli Shechtman, and Oliver Wang. The unreasonable effectiveness of deep features as a perceptual metric. In *CVPR*, pages 586–595, 2018. 6
- [81] Songchun Zhang, Yaowei Li, Junhao Zhuang, Weiyang Jin, Haoyu Wang, Xin Lu, Yilang Sun, Shiyi Zhang, Haoran Li, Xiaoxiao Ma, Yuming Li, Yijun Liu, Yaofeng Su, Yanwen Ma, Haoyu Wu, Zihan Su, Yue Ma, Lvmin Zhang, Haoyang Huang, Zeyue Xue, Anyi Rao, and Nan Duan. EchoWM: Open and enterable omnimodal world models. *arXiv preprint arXiv:2608.23189*, 2026. 2, 5, 6, 7
- [82] Jinjing Zhao, Fangyun Wei, Zhening Liu, Hongyang Zhang, Chang Xu, and Yan Lu. Spatia: Video generation with updatable spatial memory. *arXiv preprint arXiv:2512.15716*, 2025. 2, 3
- [83] Haoyi Zhu, Haozhe Liu, Yuyang Zhao, Tian Ye, Junsong Chen, Jincheng Yu, Tong He, Song Han, and Enze Xie. Sana-wm: Efficient minute-scale world modeling with hybrid linear diffusion transformer. *arXiv preprint arXiv:2605.15178*, 2026. 2, 5, 6, 7